Positive Curvature of the Spectral Trace at the Critical Line

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Abstract

Building on the exact trace formula $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ established in [5], we give a structural analysis of the second-order curvature of the spectral trace at the critical line and reduce its sign to a single weighted prime–zero sum B. The central quantity is the direct curvature sum $B = \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p)$, which satisfies the algebraic identity $O''(\frac{1}{2}) = -2B$, together with a Guinand–Weil decomposition of a smoothed prime–zero sum $\widetilde{Z}_p(\varepsilon)$ whose leading term is proved strictly negative. No hypothetical input is used: no Riemann Hypothesis, no Hilbert–Pólya postulate, no GUE conjecture.

What is proved. For every prime p and every $\varepsilon > 0$, the leading term of the Guinand–Weil decomposition satisfies $\mathrm{Main}_p(\varepsilon) < 0$ (Theorem 3.1), while the other-prime error is non-positive (Theorem 4.2). The algebraic identity $O''(\frac{1}{2}) = -2B$ connecting curvature to bias is derived from the trace formula of [5] (Proposition 6.1). The curvature sign of the spectral trace at $\sigma = \frac{1}{2}$ thus reduces to the sign of a single scalar quantity B. The decomposition $\widetilde{Z}_p(\varepsilon) = \mathrm{Main}_p + \Gamma_p + \mathrm{Const}_p + \mathrm{Err}_{p^k} + \mathrm{Err}_{\mathrm{other}}$ is the classical Guinand–Weil explicit formula applied to an even, Gaussian-damped test function; its derivation is recalled in § 2.

What is numerical. The gamma factor contribution is subleading, with $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ numerically consistent with linear scaling in ε (Theorem 4.1). The zero-sum truncation tail at $(\varepsilon, N) = (0.05, 100)$ is verified numerically to be negligible, with ratio bounded by 3×10^{-61} relative to the main term (Theorem 4.3). The proxy ratio $r^+_{p,N}(\varepsilon)$ appears unimodal on the tested ε-grid: it exceeds 1 for $p \in \{37, 53\}$ near $\varepsilon = 0.05$ and is consistent with decrease toward $r_p^\infty \le \frac{1}{2}$ as $\varepsilon \to 0$ (Theorem 5.1(a), Remark 4.2). On the sampled grid, the observed transition occurs between the tested values $\varepsilon = 0.020$ and $\varepsilon = 0.025$, with $Z^+_{p,N}(\varepsilon) < 0$ at the tested $\varepsilon = 0.020$ (ordinate proxy $Z^+_{p,N} = \Re Z_{p,\varepsilon,N}$) for every prime $p \le 53$ (Theorem 5.1(b)). The integrated bias is negative at reference parameters, $B^+_{\mathrm{int}}(0.05, 100) = -42.21 < 0$ at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ (Theorem 6.2). At the same reference parameters the direct curvature sum equals $B = -19\,342.5 < 0$, which is equivalent to $O''(\frac{1}{2}) = +38\,685 > 0$ by the curvature–bias identity (Corollary 7.1).

What is conditional. If, in addition to the numerically established inequality $B < 0$ at the reference parameters, the stationarity condition $O'(\frac{1}{2}) = 0$ holds at the same parameters, then $\sigma = \frac{1}{2}$ is a strict local minimum of O and $\mathrm{Tr}(\widetilde{T}(\sigma))$ attains a strict local maximum at $\sigma = \frac{1}{2}$ at the reference parameters (Remark following Corollary 7.1). The conditional statement is clearly marked; a weaker first-derivative cancellation estimate is formulated in Open Problem 8.1; exact stationarity $O'(\frac{1}{2}) = 0$ remains open.

What remains open. Five questions separate the present work from a full curvature proof, each precisely named in § 8. Problem 8.1 (*First-derivative cancellation*) asks for a

nontrivial cancellation estimate for the prime exponential sums $S_\kappa(\gamma_k)$ strong enough to give a genuine logarithmic saving in $|O'(\frac{1}{2})|$ relative to the trivial triangle bound. Problem 8.2 (*Asymptotic pointwise bias*) asks for an analytic proof that $Z_p^\infty(\varepsilon) < 0$ for every prime $p \le \kappa$ and sufficiently small $\varepsilon > 0$, together with a finite-N transfer to $Z_{p,N}^+(\varepsilon)$. Problem 8.3 (*Uniformity of positive curvature*) asks whether the inequalities $B < 0$ and $O''(\frac{1}{2}) > 0$ persist at parameter points (κ, ε, N) other than the reference values tested here. Problem 8.4 (*Integrated-to-direct transfer*) asks whether pointwise or integrated bias implies the negativity of B. Problem 8.5 (*Lagarias–Weil bridge*) asks whether the curvature statement transfers to the Weil-positivity framework of [7].

Structure. The formal core in §§ 2–7 is non-circular; analytic identities, numerical observations, and conditional statements are explicitly separated. No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–7.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The trace parameter $\sigma \in \mathbb{R}$ is evaluated at $\sigma = \frac{1}{2}$ unless stated otherwise.

Global parameters. The reference values are $\kappa = 53$ (prime cutoff, yielding the 16 active primes $p \le \kappa$), $\varepsilon = 0.05$ (Gaussian regularisation parameter), and $N = 100$ (number of Riemann ordinates γ_k employed). These values are fixed throughout §§ 2–7 unless explicitly varied. The associated spectral weight sums are

$$W_{\varepsilon,N} := \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}, \qquad P(\kappa) := \sum_{p\le\kappa} \log p.$$

Main objects. The finite truncation

$$Z_p \;=\; Z_{p,\varepsilon,N} \;:=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k}$$

is the numerical object of [4, 5]. The γ_k^2-weighted variant

$$Z_p^{(2)} \;:=\; \sum_{k=1}^{N} \gamma_k^2\, e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k}$$

carries the curvature weighting needed for the bias decomposition; the dual form of the curvature sum is $B = \sum_p (\log p)^2 \operatorname{Re} Z_p^{(2)}$. The associated Guinand–Weil object

$$\widetilde{Z}_p(\varepsilon) \;:=\; \sum_{\rho} h_{p,\varepsilon}^{\text{even}}\left(\frac{\rho-1/2}{i}\right), \qquad h_{p,\varepsilon}^{\text{even}}(u) = e^{-\varepsilon^2 u^2}\cos(u\log p),$$

sums over all non-trivial zeros of ζ. $\widetilde{Z}_p(\varepsilon)$ is real-valued by the quartet symmetry $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$ of the non-trivial zeros. The finite positive-ordinate cosine proxies $Z^+_{p,N}(\varepsilon) = \Re Z_{p,\varepsilon,N}$ and $\Re Z^{(2)}_p$ are real by construction (cosine kernel). The phasor sums Z_p and $Z^{(2)}_p$ themselves are complex; only their real parts enter the cosine-kernel computations used here. The identity $\widetilde{Z}^{\mathrm{full}}_p = 2\,Z^+_p$ holds when all non-trivial zeros lie on the critical line; in general $\widetilde{Z}^{\mathrm{full}}_p = 2\,Z^+_p + E^{\mathrm{off}}_p$ where E^{off}_p captures off-critical-line contributions (see Paper 5, Open Problem 6.5 (GW bridge)). All numerical computations in this paper use the ordinate proxy $Z^+_p(\varepsilon) = \Re Z_{p,\varepsilon,N}$ directly. The Guinand–Weil object $\widetilde{Z}_p$ motivates the decomposition but the actual numerical values refer to Z^+_p. At $(\varepsilon, N) = (0.05, 100)$ the truncation tail $R_{p,N}$ is negligible by Theorem 4.3. The direct curvature sum

$$B \;:=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p)$$

enters the algebraic identity $O''(\tfrac{1}{2}) = -2B$ (Proposition 6.1); the integrated bias is $B^+_{\mathrm{int}}(\varepsilon, N) \;:=\; \sum_p (\log p)^2\,Z^+_{p,N}(\varepsilon)$.

Analytical foundations. The identity $O''(\tfrac{1}{2}) = -2B$ (Proposition 6.1) is obtained by direct differentiation of the trace oscillatory sum of Paper 5. The Guinand–Weil decomposition in § 2 uses the classical explicit formula framework of Weil [9] and Guinand [8]. The truncation estimate in Theorem 4.3 uses Riemann–von Mangoldt zero-counting estimates [20, Thm. 5.8] and Gaussian tail bounds.

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of working precision; Riemann zeros γ_k were evaluated via `mpmath.zetazero`. Verification code is available at [6] (`https://github.com/utehrani/analysislab-nt`).

Not to be confused with. B (curvature sum, weight $(\gamma_k \log p)^2$) and B_{int} (integrated bias, weight $(\log p)^2$) differ in their weighting and in the object summed; both are negative at the reference parameters, but neither implies the other (Open Problem 8.4). Z_p (finite truncation, level N, sum over positive ordinates $k = 1, \ldots, N$, in contract notation Z^+_p), $Z^{(2)}_p$ (with γ_k^2-weighting, used in B), and $\widetilde{Z}_p$ (Guinand–Weil object, sum over all non-trivial zeros, in contract notation $\widetilde{Z}^{\mathrm{full}}_p$) are three formally distinct objects. For h^{even} and $N \to \infty$, the identity $\widetilde{Z}^{\mathrm{full}}_p(\varepsilon) = 2\,Z^+_p(\varepsilon)$ holds conditionally on all non-trivial zeros being on the critical line; without this assumption, an off-critical contribution E^{off}_p must be added (cf. Paper 5, Open Problem 6.5 (GW bridge)). Numerical computations therefore refer to the ordinate proxy $Z^+_p(\varepsilon)$, not the full GW object $\widetilde{Z}^{\mathrm{full}}_p$. $Z^{(2)}_p$ carries the additional spectral weight absent from Z_p. The diagonal-energy constants $D \approx 9.470$ of [2] and $D_{\mathrm{SEL}} \approx 10.985$ of [5] share a name across the series but are defined in different operator contexts and must not be identified.

1 Introduction

1.1 Background and Motivation

This paper is the sixth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The series is built around a finite-dimensional Hilbert-space model in which prime numbers mediate coupling between zeros, encoded in an explicit operator family.

Paper 1 [1] introduced the local curvature function $H_{\mathrm{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \le \kappa} V_p(\sigma)$, identifying $\sigma = \frac{1}{2}$ as a phase boundary through the divergence $H_{\mathrm{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2$.

Paper 2 [2] connected this local curvature to the Weil explicit functional, constructing admissible test functions for which the curvature divergence becomes a structural feature of the Weil energy.

Paper 3 [3] made the geometric structure explicit: two finite-dimensional Hilbert spaces H_{str} (over primes) and H_{null} (over zeros) connected by a linear coupling map Φ, with loop operator $T = \Phi^* \circ \Phi$ and a first numerical verification of the energy asymmetry $\eta_{\mathrm{orig}} > 0$.

Paper 4 [4] introduced the dual operator $\widetilde{T} = \Phi \circ \Phi^*$ on H_{null} — the Tehrani operator — and established a conditional proof of energy asymmetry under the remainder-control assumption (E_{rem}), motivated by prime-phase cancellation estimates.

Paper 5 [5] extended this to a one-parameter family $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ and proved the exact spectral trace identity

$$\mathrm{Tr}(\widetilde{T}(\sigma)) \;=\; D_{\mathrm{SEL}} - O(\sigma), \qquad O(\sigma) \;=\; \frac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p),$$

where D_{SEL} is a σ-independent constant. Four independent numerical signatures concentrating at $\sigma = \frac{1}{2}$ were documented in [4, 5], and the curvature-bias quantity $B = \sum_p (\log p)^2 \, \mathrm{Re}(Z_p^{(2)})$ was introduced with the reference value $B = -19\,342.5$ at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$.

The present paper addresses the analytic question left open by [5]: *is there a structural reason, independent of the Riemann Hypothesis, why $\sigma = \frac{1}{2}$ is distinguished by the trace?* We reduce this question at second order to the sign of the direct curvature sum B via the algebraic identity $O''(\frac{1}{2}) = -2B$, and verify the positive-curvature sign numerically at the reference parameters. At the reference parameters B is numerically negative, hence $O''(\frac{1}{2})$ is numerically positive. The paper establishes this curvature statement together with a Guinand–Weil decomposition of a related smoothed prime–zero sum, recalls four further numerical signatures from Papers 4–5 concentrating at $\sigma = \frac{1}{2}$, and formulates five open problems constituting the completion path toward a theorem-level selection principle.

The operator $\widetilde{T}(\sigma)$ should be viewed neither as a Hilbert–Pólya spectral realisation in the sense of Connes [12] or Berry–Keating [13], nor as a de Branges structure-function criterion [14]; it is a finite-cutoff operator packaging Gaussian-smoothed explicit-formula data, closest in spirit to Hilbert-space reformulations of the explicit formula such as those of Burnol [15].

The heuristic labels "prime field" and "field strength" used in the series denote a finite-cutoff coupling picture in which primes mediate pairwise interactions between zero ordinates via explicit Gaussian-smoothed phasors; this should be distinguished from the periodic-orbit dictionary of Berry–Keating [13], the adelic trace-formula picture of Connes [12], and the Frobenius/cohomology picture of function-field zeta functions.

1.2 Main Results

The paper contains three main components.

The first is the curvature statement at the critical line. At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, direct numerical evaluation gives $B = -19\,342.5 < 0$, which is equivalent via the algebraic identity of Proposition 6.1 to $O''(\frac{1}{2}) = -2B = +38\,685 > 0$ (Corollary 7.1). A strict local minimum of O at $\sigma = \frac{1}{2}$ would require additionally that $O'(\frac{1}{2}) = 0$. At the reference parameters, $O'(\frac{1}{2}) = +2.475 \neq 0$, so $\sigma = \frac{1}{2}$ is not an exact local minimum of O at the reference finite model. A sufficiently small stationarity bound, together with a local convexity lower bound for O'' near $\sigma = \frac{1}{2}$, would locate a local minimum near $\sigma = \frac{1}{2}$ (Problem 8.1).

The curvature identity $O''(\frac{1}{2}) = -2B$ is structurally analogous to the Li/Bombieri–Lagarias sign criteria [17, 18] in that it turns explicit-formula data into a sign condition, but it is not a derivative of ξ at $s = 1$ and not a Weil-positivity theorem; it is a local second-order statement in the auxiliary trace parameter σ at the symmetry point $\frac{1}{2}$.

Bombieri [16] studies the positivity of the Weil quadratic functional $Q(g) = W(g * \tilde{g})$ truncated to $L^2[-t, t]$, proving positivity for small t and showing that off-critical zeros contribute exactly half a negative eigenvalue each. The present paper does not truncate the functional in a spatial variable: our operator $\widetilde{T}(\sigma) = \Phi(\sigma)\Phi(\sigma)^*$ is a Gram-type finite-rank matrix indexed by explicit Gaussian-damped prime phasors at spectral cutoff κ and regularisation ε, and our central object is the *second derivative* $O''(\frac{1}{2}) = -2B$ of an exact trace (not a spectral positivity condition on Q). The Bombieri quadratic form is the appropriate ambient framework for a possible future construction of the Weil transfer operator $W_{\kappa,\varepsilon}$.

The second is the Guinand–Weil bias analysis of §§ 2–5. The smoothed prime–zero sum $\widetilde{Z}_p(\varepsilon)$ is decomposed into five contributions, of which the leading one is proved negative (Theorem 3.1), and the other-prime contribution is proved non-positive (Theorem 4.2), while the zero-ordinate truncation tail is numerically verified, with analytic tail control, to be negligible at the reference parameters (Theorem 4.3). The gamma contribution is shown numerically to be subleading of order ε (Theorem 4.1). The proxy ratio $r^+_{p,N}(\varepsilon)$ is unimodal on the tested grid (Theorem 5.1(a), Remark 4.2), and a sign-crossover is localised in $(0.020, 0.025)$ (Theorem 5.1(b)). The integrated bias $B^+_{\mathrm{int}}(\varepsilon, N) = \sum_{p \le \kappa} (\log p)^2\, Z^+_{p,N}(\varepsilon)$ is numerically negative at the reference parameters, with $B^+_{\mathrm{int}}(0.05, 100) = -42.21$ (Theorem 6.2).

The third is the structural separation made explicit in § 6. The quantities B, $B_{\mathrm{int}}(\varepsilon)$, and the pointwise sign of $Z^+_{p,N}(\varepsilon)$ are structurally distinct witnesses of the same qualitative phenomenon: B carries the $(\gamma_k \log p)^2$-weighting, B_{int} only the $(\log p)^2$-weighting, and the pointwise statement carries no weighting at all. B and B^+_{int} are numerically negative at the reference parameters, and the pointwise bias $Z^{+}_{p,N}(0.05) < 0$ holds for 14 of the 16 primes; the analytical relationship between them is left open as Problem 8.4. The five open problems of § 8 list the steps remaining, in order of what we consider tractability: Problem 8.1 (first-derivative cancellation) is the most tractable and, together with the numerically established $B < 0$ and local convexity, would locate a strict local minimum of O near $\sigma = \frac{1}{2}$. An exact strict local minimum at $\sigma = \frac{1}{2}$ requires the stronger condition $O'(\frac{1}{2}) = 0$.

1.3 Reader Contract

This paper *proves*, by direct computation and classical estimates of the Guinand–Weil type, that the main term is negative (Theorem 3.1), that the other-prime error is non-positive (Theorem 4.2), and the algebraic curvature–bias identity $O''(\frac{1}{2}) = -2B$ (Proposition 6.1). These are unconditional mathematical statements.

This paper *establishes numerically*, at the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, that the truncation tail is negligible at the reference parameters (Theorem 4.3, numerically verified at $(\varepsilon, N) = (0.05, 100)$), that the gamma contribution is subleading of order ε (Theorem 4.1), that the proxy ratio $r^{+}_{p,N}(\varepsilon)$ is unimodal on the tested grid and is consistent with decrease toward $r^{\infty}_{p} \leq \frac{1}{2}$ (Theorem 5.1(a), Remark 4.2), and that the pointwise bias $Z^{+}_{p,N}(\varepsilon) < 0$ holds on the tested subgrid at $\varepsilon = 0.020$ (via ordinate proxy; Theorem 5.1), that $B^{+}_{\text{int}}(0.05, 100) = -42.21 < 0$ (Theorem 6.2), and that the positive-curvature statement $O''(\frac{1}{2}) = +38\,685 > 0$ holds at the reference parameters (Corollary 7.1). These statements are verified on a finite grid and are clearly labelled [numerical] in their theorem headings.

This paper *states conditionally* that exact stationarity $O'(\frac{1}{2}) = 0$, together with the numerically established $O''(\frac{1}{2}) > 0$ at the reference parameters, would make $\sigma = \frac{1}{2}$ a strict local minimum of O (Remark following Corollary 7.1). A stationarity bound $|O'(\frac{1}{2})| \ll W_{\varepsilon,N} \cdot P(\kappa)$ would become a localisation statement only when additionally combined with a local convexity lower bound and the smallness condition stated in Problem 8.1. No part of the unconditional claims above depends on either condition.

This paper *leaves open* five well-posed questions named in § 8: Problem 8.1 (First-derivative cancellation), Problem 8.2 (Asymptotic pointwise bias), Problem 8.3 (Uniformity of positive curvature), Problem 8.4 (Integrated-to-direct transfer), and Problem 8.5 (Lagarias–Weil bridge). The first four are formulated entirely within classical analytic number theory; the fifth names the remaining bridge to Weil positivity in the formulation of [7].

2 The Guinand–Weil Decomposition

Throughout, p denotes a rational prime, $\varepsilon > 0$ a smoothing parameter, and $\rho = \beta + i\gamma$, $0 < \beta < 1$, a non-trivial zero of the Riemann zeta function. For critical-line zeros, $(\rho - \frac{1}{2})/i = \gamma$; off the critical line, the argument is complex. The real-valuedness of $\widetilde{Z}_p$ follows from the quartet symmetry $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$ of the non-trivial zeros. We fix a cutoff $\kappa \geq 2$ and consider only primes $p \leq \kappa$. For numerical reference we use $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, where N is the number of zeros used in the finite approximation; the first 16 primes lie below $\kappa = 53$. The Guinand–Weil explicit formula (see e.g. [20, § 5.5]) is the key analytic tool; its admissibility conditions on the test function are recalled below.

Definition 2.1 (Smoothed zero sum)**.** *Let*

$$Z_p \;=\; Z_{p,\varepsilon,N} \;:=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k}, \qquad \operatorname{Re} Z_p \;=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2} \cos(\gamma_k \log p).$$

This is the finite numerical object studied in [4, 5]. The associated object obtained from the Guinand–Weil explicit formula is

$$\widetilde{Z}_p(\varepsilon) \;:=\; \sum_{\rho} h^{\mathrm{even}}_{p,\varepsilon}\!\left(\frac{\rho-\frac{1}{2}}{i}\right), \qquad h^{\mathrm{even}}_{p,\varepsilon}(u) \;=\; e^{-\varepsilon^2u^2}\cos(u\log p).$$

The Gaussian $h^{\mathrm{even}}_{p,\varepsilon}$ is admissible in the sense of Guinand–Weil (entire of order 2, Schwartz-class decay, even, real-valued; cf. Carneiro–Littmann–Vaaler [22] for the Beurling–Selberg Gaussian-subordination framework; and Carneiro–Milinovich–Soundararajan [23] for Fourier-optimization methods in the explicit-formula context). Since $h^{\mathrm{even}}_{p,\varepsilon}$ is even and decays rapidly, $\widetilde{Z}_p(\varepsilon)$ is real-valued and absolutely convergent. Its Fourier transform is

$$\widehat{h}_{p,\varepsilon}(x) \;=\; \frac{\sqrt{\pi}}{2\varepsilon}\left[\exp\!\left(-\tfrac{(\log p-2\pi x)^2}{4\varepsilon^2}\right) + \exp\!\left(-\tfrac{(\log p+2\pi x)^2}{4\varepsilon^2}\right)\right],$$

which is strictly positive on $\mathbb{R}$ and concentrates at $\pm\log p/(2\pi)$ as $\varepsilon \to 0$. We use the Fourier convention $\widehat{h}(x) = \int_{-\infty}^{\infty} h(u)\, e^{-2\pi i u x}\, du$, matching the arguments $\log n/(2\pi)$ in the Guinand–Weil formula.

Proposition 2.2 (Guinand–Weil decomposition convention; classical formula recalled)**.** *For every prime p and every $\varepsilon > 0$,*

$$\widetilde{Z}_p(\varepsilon) \;=\; \mathrm{Main}_p(\varepsilon) + \Gamma_p(\varepsilon) + \mathrm{Const}_p(\varepsilon) + \mathrm{Err}_{p^k}(\varepsilon) + \mathrm{Err}_{\mathrm{other}}(\varepsilon),$$

where the five contributions are defined by the standard partition of the prime sum and the archimedean terms of the explicit formula. Explicitly,

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{1}{2\pi}\cdot\frac{\log p}{\sqrt{p}}\left[\widehat{h}_{p,\varepsilon}\!\left(\tfrac{\log p}{2\pi}\right) + \widehat{h}_{p,\varepsilon}\!\left(-\tfrac{\log p}{2\pi}\right)\right],$$

$$\Gamma_p(\varepsilon) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\varepsilon^2t^2}\cos(t\log p)\,\operatorname{Re}\psi\!\left(\tfrac{1}{4}+\tfrac{it}{2}\right)\,dt,$$

$$\mathrm{Const}_p(\varepsilon) \;=\; \widetilde{Z}_p(\varepsilon) \;-\; \mathrm{Main}_p(\varepsilon) \;-\; \Gamma_p(\varepsilon) \;-\; \mathrm{Err}_{p^k}(\varepsilon) \;-\; \mathrm{Err}_{\mathrm{other}}(\varepsilon),$$

where ψ denotes the digamma function. The prime-power error Err_{p^k} *collects the contribution from $n = p^m$, $m \geq 2$ (higher powers of the fixed prime p). The remaining-prime error* $\mathrm{Err}_{\mathrm{other}}$ *collects the contribution from $n = q^m$ with $q \neq p$ prime and $m \geq 1$ (all prime powers not involving p). Both are standard sums $\sum_n \Lambda(n)\widehat{h}_{p,\varepsilon}(\log n/(2\pi))/\sqrt{n}$.*

This proposition records the decomposition convention induced by the classical Guinand–Weil explicit formula. The present paper does not supply a new proof of the explicit formula; it invokes the standard formula with the stated Fourier normalisation and defines Const_p as the residual after removing the other displayed contributions.

Remark. *At the test-function evaluations corresponding to the zeta singularity at* $s = 1$ *and its functional-equation reflection at* $s = 0$*, the completed explicit formula contributes the value* $h^{\mathrm{even}}_{p,\varepsilon}(u)$ *evaluated at* $u = \pm i/2$*. A direct computation gives*

$$h^{\mathrm{even}}_{p,\varepsilon}\big(\tfrac{i}{2}\big) + h^{\mathrm{even}}_{p,\varepsilon}\big(-\tfrac{i}{2}\big) \;=\; 2\, e^{\varepsilon^2/4} \cosh\Big(\tfrac{\log p}{2}\Big) \;=\; e^{\varepsilon^2/4}\Big(\sqrt{p} + \tfrac{1}{\sqrt{p}}\Big).$$

This is the hyperbolic cosine, not the circular cosine: the formal substitution $\cos(i \log p/2) = \cosh(\log p/2)$ *is essential. Replacing* cosh *by* cos *would yield a quantity that is negative for* $p \geq 29$*, which is analytically impossible since the left-hand side equals* $\sqrt{p} + 1/\sqrt{p} > 2$ *for every* $p \geq 2$*. The correct* cosh *form is used throughout what follows. See Edwards [11, §1.8, §1.16, and §3.6] for the classical derivation of the pole contributions at* $s = 0, 1$*.*

The remainder of the paper studies the five terms of the decomposition in turn.

3 The Main Term

Theorem 3.1 (Main term negativity; proved)**.** *For every prime* p *and every* $\varepsilon > 0$*,*

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{\log p}{2\sqrt{\pi}\,\varepsilon\,\sqrt{p}} \cdot \big(1 + e^{-(\log p)^2/\varepsilon^2}\big) \;<\; 0.$$

Proof. The Fourier transform $\widehat{h}_{p,\varepsilon}(x)$ is strictly positive on $\mathbb{R}$. The two evaluations at $x = \log p/(2\pi)$ and $x = -\log p/(2\pi)$ sum to $(\sqrt{\pi}/\varepsilon)(1 + e^{-(\log p)^2/\varepsilon^2})$, with the reflected term exponentially suppressed: at $\varepsilon = 0.05$ and $p = 2$ it is below 10^{-82}, and decreases further for larger p. Collecting the dominant contribution,

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{1}{2\pi} \cdot \frac{\log p}{\sqrt{p}} \cdot \frac{\sqrt{\pi}}{\varepsilon}\big(1 + e^{-(\log p)^2/\varepsilon^2}\big) = -\frac{\log p}{2\sqrt{\pi}\,\varepsilon\,\sqrt{p}}\big(1 + e^{-(\log p)^2/\varepsilon^2}\big).$$

As $\varepsilon \to 0$, the factor $e^{-(\log p)^2/\varepsilon^2} \to 0$, giving $|\mathrm{Main}_p(\varepsilon)| \sim (\log p)/(2\sqrt{\pi}\,\varepsilon\,\sqrt{p})$. The sign is negative because all factors in the leading expression are positive and enter with a minus sign. □

Corollary 3.2 (Main-term scaling; proved)**.** *As* $\varepsilon \to 0$*,*

$$|\mathrm{Main}_p(\varepsilon)| \sim \frac{\log p}{2\sqrt{\pi}\,\varepsilon\sqrt{p}},$$

giving a leading-order scaling of $1/\varepsilon$ *with magnitude constant* $1/(2\sqrt{\pi}) \approx 0.2821$*; the signed main term carries an additional minus sign from the prime-sum formula.*

At the reference parameter $\varepsilon = 0.05$ and for $2 \le p \le 53$, the main term takes values in the interval $[-4.15, -2.77]$, minimised at $p = 7$ and maximised at $p = 2$.

4 Error Terms

We now treat the three remaining contributions other than the constant term.

4.1 The gamma contribution

Theorem 4.1 (Gamma term is subleading; numerical)**.** *On the tested finite grid $p \le 53$, $\varepsilon \in \{0.005, 0.010, \ldots, 0.100\}$, the ratio $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is numerically consistent with linear scaling in ε; a log–log regression gives slope 1.001. At $\varepsilon = 0.05$, the ratio lies in $[0.017, 0.172]$ for all tested primes ($p \le 53$); in particular $|\Gamma_p(0.05)| < |\mathrm{Main}_p(0.05)|$ for all 16 primes $p \le 53$.*

Numerical verification. For every fixed $\varepsilon > 0$, the Gaussian factor $e^{-\varepsilon^2 t^2}$ makes the integrand absolutely integrable on $\mathbb{R}$. The claimed ratio range and log–log slope are established by direct numerical quadrature over the grid $p \le 53$, $\varepsilon \in \{0.005, 0.010, \ldots, 0.100\}$ (step 0.005, giving 20 values), pooling all 16 primes; the slope is the least-squares slope of $\log(|\Gamma_p|/|\mathrm{Main}_p|)$ against $\log \varepsilon$. No limit $\varepsilon \to 0$ is asserted. Establishing an Abel-regularised bound uniform in p, which would require explicit control of the oscillatory integral as $\varepsilon \downarrow 0$, is part of Problem 8.2. ☐

4.2 The prime-power and other-prime errors

Theorem 4.2 (Other-prime error is non-positive; proved)**.** *For every fixed prime $p \le \kappa$ and every $\varepsilon > 0$,*

$$\mathrm{Err}_{\mathrm{other},p}(\varepsilon) \;\le\; 0.$$

Proof. By the standard form of the explicit formula,

$$\mathrm{Err}_{\mathrm{other}}(\varepsilon) \;=\; -\frac{1}{2\pi} \sum_{\substack{q \le \kappa \\ q \ne p \\ q \text{ prime}}} \frac{\log q}{\sqrt{q}} \left[\widehat{h}_{p,\varepsilon}\!\left(\tfrac{\log q}{2\pi}\right) + \widehat{h}_{p,\varepsilon}\!\left(-\tfrac{\log q}{2\pi}\right)\right] + (\text{higher-prime-power terms}).$$

Each summand has $\log q > 0$, $\sqrt{q} > 0$, and $\widehat{h}_{p,\varepsilon}(\cdot) > 0$ on $\mathbb{R}$, with an overall minus sign. The higher-prime-power terms share the same sign structure with $\Lambda(q^k) = \log q$ and the same positive transform. Hence the full sum is non-positive. The identical sign argument applies to all primes $q > \kappa$ and higher prime powers with $q \ne p$: the positivity of $\widehat{h}_{p,\varepsilon}$ on $\mathbb{R}$ and the negativity of the overall sign extend without restriction to the full prime-power range in the explicit formula. Hence $\mathrm{Err}_{\mathrm{other}}(\varepsilon) \le 0$ for the complete GW sum. ☐

Theorem 4.2 is a structural identity rather than an error bound: the contribution from primes $q \ne p$ actively reinforces the negativity of $\widetilde{Z}_p$.

Theorem 4.3 (Truncation error is negligible; numerical)**.**

$$\frac{|R_{p,N}(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|} \;\le\; 3\times 10^{-61}$$

at the reference parameters $(\varepsilon, N) = (0.05, 100)$ *for every prime* $p \le 53$*, where* $R_{p,N}(\varepsilon) = \sum_{k>N} e^{-\varepsilon^2\gamma_k^2}\cos(\gamma_k \log p)$ *is the tail of the zero-ordinate sum.*

Proof. The triangle inequality gives $|R_{p,N}(\varepsilon)| \le \sum_{k>N} e^{-\varepsilon^2\gamma_k^2}$. We bound this tail by partial summation against the Riemann–von Mangoldt density estimate $N(t+1) - N(t) \ll \log(t+2)$ ([20, Thm. 5.8], with explicit zero-counting bounds from Bellotti–Wong [30]):

$$\sum_{\gamma>\gamma_{100}} e^{-\varepsilon^2\gamma^2} \;\ll\; e^{-\varepsilon^2\gamma_{100}^2}\log(\gamma_{100}+2) + 2\varepsilon^2\int_{\gamma_{100}}^{\infty} e^{-\varepsilon^2 t^2}\, t\,\log(t+2)\,dt.$$

At $(\varepsilon, N) = (0.05, 100)$, $\gamma_{100} = 236.524$, the Gaussian factor $e^{-\varepsilon^2\gamma_{100}^2} = e^{-0.0025\cdot 236.524^2} \approx e^{-140}$ renders both terms negligible. A standard Gaussian tail bound gives $\int_{236}^{\infty} e^{-0.0025t^2}\log t\,dt \ll e^{-140}$, and the lower bound $|\mathrm{Main}_p(\varepsilon)| \ge \log 2/(2\sqrt{\pi}\cdot 0.05\cdot\sqrt{53})$ over the tested prime range yields $|R_{p,N}|/|\mathrm{Main}_p| \le 3\times 10^{-61}$. Accurate evaluation of the partial sums follows Brent–Platt–Trudgian [31]. For the γ_k^2-weighted tail: the function $t^2 e^{-\varepsilon^2 t^2}$ is monotone decreasing for $t > 1/\varepsilon = 20$, and $\gamma_{100} = 236.524 \gg 1/\varepsilon$. Hence for $k > N$, $\gamma_k > \gamma_{100}$ and $\gamma_k^2 e^{-\varepsilon^2\gamma_k^2} \le \gamma_{100}^2 e^{-\varepsilon^2\gamma_{100}^2} \approx \gamma_{100}^2\cdot e^{-140}$. By partial summation the sum satisfies

$$\sum_{k>N}\gamma_k^2 e^{-\varepsilon^2\gamma_k^2} \;\ll\; \gamma_{100}^2\, e^{-\varepsilon^2\gamma_{100}^2}\log(\gamma_{100}+2) + \int_{\gamma_{100}}^{\infty}(2t+2\varepsilon^2 t^3)e^{-\varepsilon^2 t^2}\log(t+2)\,dt \;\ll\; \gamma_{100}^2\cdot e^{-140},$$

which is negligible relative to $|B|$ at the reference parameters. The numerical ratio $|R_{p,N}|/|\mathrm{Main}_p| \le 3\times 10^{-61}$ is verified directly by `mpmath` computation at 50-digit precision in the accompanying script [6]: the sum $\sum_{k=101}^{1000} e^{-\varepsilon^2\gamma_k^2}$ is evaluated numerically, confirming the stated bound for all primes $p \le 53$ at the reference parameters. The remaining infinite tail $k > 1000$ is bounded analytically. At $\gamma_{1000} \approx 1419$ and $\varepsilon = 0.05$, the Riemann–von Mangoldt estimate gives

$$\sum_{k>1000} e^{-\varepsilon^2\gamma_k^2} \;\ll\; e^{-\varepsilon^2\gamma_{1000}^2}\log(\gamma_{1000}+2) \;\approx\; e^{-5034}\cdot 7.3 \;\approx\; 10^{-2184},$$

which is negligible relative to the partial sum above and to the stated bound 3×10^{-61}. The full infinite-tail control used for the numerical verification is obtained by combining the numerically computed finite partial sum ($k = 101, \ldots, 1000$) with this explicit analytic estimate. □

For the ordinate proxy $Z_{p,N}^{+}$ at the reference parameters, the $N = 100$ unweighted truncation tail is negligible relative to $|\mathrm{Main}_p|$ (Theorem 4.3). The analogous γ_k^2-weighted tail entering B is controlled by the same Gaussian mechanism; no separate quantified bound is stated here, as the sign of $B = -19\,342.5$ is unaffected by the tail at the stated numerical precision.

Remark (Asymptotic constant-term ratio; numerical)**.** *The proxy ratio* $r_{p,N}^{+}(\varepsilon) := |\mathrm{Const}_{p,N}^{+}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ *is observed, on the smallest tested* ε*-values, to be consistent with decrease toward a limiting value as* $\varepsilon \downarrow 0$*. At the smallest tested values (*$\varepsilon \le 0.015$*), all ratios lie below* 0.82 *for every* *prime $p \le 53$, consistent with convergence to a limit $r_p^\infty \le \frac{1}{2}$ as $\varepsilon \to 0$. An analytic proof of $r_p^\infty = \frac{1}{2}$ requires an explicit asymptotic expansion of* $\mathrm{Const}_p(\varepsilon)$ *as $\varepsilon \to 0$ beyond the leading $e^{\varepsilon^2/4}\cosh(\frac{\log p}{2})$ pole term evaluated in § 2; this is part of Problem 8.2. On the smallest tested values, the data are consistent with* $\mathrm{Const}_p(\varepsilon) + \mathrm{Main}_p(\varepsilon)$ *decreasing toward zero; no asymptotic proof is supplied here.*

5 The Constant Term and the Sign-Crossover Localisation

The constant term $\mathrm{Const}_p(\varepsilon)$ is the only contribution whose full asymptotic structure is not explicitly resolved in the present work. It collects the $s = 0, 1$ endpoint contributions in the completed Guinand–Weil formula (associated with the zeta singularity at $s = 1$ and its reflection at $s = 0$ under the functional equation), the $\log(2\pi)$ normalisation, and the remaining endpoint and normalisation terms not included in the displayed prime-power sums. It is the only term of the decomposition that resists a short analytic treatment. We state a numerical result on its asymptotic ratio to the main term and, independently, a numerical result localising the sign-crossover of $Z^+_{p,N}(\varepsilon)$ on a finite ε-grid.

Theorem 5.1 (Finite-grid proxy-ratio pattern and sign-crossover localisation; numerical)**.** *Numerically, the following hold.*

(a) *The proxy ratio $r^+_{p,N}(\varepsilon) := |\mathrm{Const}^+_{p,N}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ appears unimodal on the tested ε-grid, with an observed maximum near the largest tested value $\varepsilon = 0.050$; no continuous unimodality statement is proved. The maximum exceeds 1 for $p = 37$ and $p = 53$ (attaining 1.17 and 1.21 respectively at $\varepsilon = 0.05$). On the smallest tested values $\varepsilon \in \{0.005, 0.010, 0.015\}$, all ratios lie below 0.82 for every prime $p \le 53$. The numerical data are consistent with decrease toward $r_p^\infty \le \frac{1}{2}$ for the underlying analytical object; no analytic limit is proved (Remark 4.2).*

(b) *On the sampled grid, the bias changes sign between the tested values $\varepsilon = 0.020$ and $\varepsilon = 0.025$. At the tested value $\varepsilon = 0.020$, one finds*

$$Z^+_{p,N}(\varepsilon) \; < \; 0 \qquad \textit{for every prime } p \le 53 \quad (\textit{ordinate proxy } Z^+_{p,N} = \Re Z_{p,\varepsilon,N}).$$

Both statements are numerical and refer to the cutoff $\kappa = 53$. The location of the sign-crossover and the sign at smaller tested values of ε are empirical findings on a finite grid; no continuous interval-persistence statement is asserted. The analytical question behind (a) is documented as Problem 8.2 of § 8.

Discussion. The $s = 1$ endpoint evaluation gives $h^{\mathrm{even}}_{p,\varepsilon}(i/2) + h^{\mathrm{even}}_{p,\varepsilon}(-i/2) = e^{\varepsilon^2/4}(\sqrt{p} + 1/\sqrt{p})$, which is $O(1)$ as $\varepsilon \downarrow 0$ for fixed p (see Edwards [11, §1.8, §1.16, and §3.6]). The observed ε^{-1} component of the proxy residual $\mathrm{Const}^+_{p,N}$ arises from the full residual in the chosen partition and is not an analytic property of the endpoint term alone; an explicit derivation is part of Problem 8.2. The limiting coefficient of the proxy ratio is not produced by the endpoint term alone. It arises from the positive-ordinate proxy normalisation: under the GW bridge, $Z^+_{p,N} \approx \frac{1}{2}\widetilde{Z}^{\mathrm{full}}_p$, and hence $\mathrm{Const}^+_{p,N} \approx \frac{1}{2}\mathrm{Const}_p - \frac{1}{2}(\mathrm{Main}_p + \Gamma_p + \mathrm{Err}_{p^k} + \mathrm{Err}_{\mathrm{other}})$. Since $\mathrm{Main}_p(\varepsilon) \sim -\frac{\log p}{2\sqrt{\pi}\sqrt{p}} \cdot \varepsilon^{-1}$, this identity explains the observed leading contribution $\mathrm{Const}^+_{p,N} \approx -\frac{1}{2}\mathrm{Main}_p$ and hence the expected ratio $r_p^\infty \to \frac{1}{2}$, modulo the open GW bridge and analytic control of the remaining terms. Numerical evaluation of $\mathrm{Const}^+_{p,N}(\varepsilon)$ for $\varepsilon \in \{0.005, 0.010, 0.015, 0.020, 0.025, 0.030, 0.050\}$ across all primes $p \le 53$ shows that the proxy ratio $|\mathrm{Const}^+_{p,N}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is a function whose values on the tested grid exhibit a single peak near $\varepsilon = 0.050$ and limiting value below 0.82 on the smallest tested grid values ($\varepsilon \le 0.015$), consistent with decrease toward a limiting value $r_p^\infty \le \frac{1}{2}$ (Remark 4.2). For $p \in \{37, 53\}$ this maximum narrowly exceeds unity (attaining 1.183 and 1.216 respectively), which produces two pointwise exceptions to the bias at $\varepsilon = 0.05$; see below. The sign-crossover localisation in (b) is obtained from the grid computation directly, not as an analytic consequence of (a).

Numerical evidence. At $\varepsilon = 0.05$ and $\kappa = 53$, the proxy-residual values satisfy $\mathrm{Const}^+_{p,N} \in [2.44, 3.93]$ with mean 3.27 and standard deviation 0.46 (coefficient of variation 14%). The displayed $\mathrm{Const}^+_{p,N}(\varepsilon)$ values are computed as the residual $\mathrm{Const}^+_{p,N}(\varepsilon) := Z^+_{p,N}(\varepsilon) - \mathrm{Main}_p(\varepsilon) - \Gamma_p(\varepsilon) - \mathrm{Err}_{p^k}(\varepsilon) - \mathrm{Err}_{\mathrm{other}}(\varepsilon)$ at the reference parameters $(\varepsilon, N) = (0.05, 100)$. These numerical values are the ordinate-proxy residuals: the left-hand side is the positive-ordinate half-sum $Z^+_{p,N}$, while the subtracted terms $(\mathrm{Main}_p, \Gamma_p, \mathrm{Err}_{p^k}, \mathrm{Err}_{\mathrm{other}})$ are the full Guinand–Weil terms of Proposition 2.2. Since $Z^+_{p,N} \approx \frac{1}{2}\widetilde{Z}_p$ under the bridge, a short computation gives

$$\mathrm{Const}^+_{p,N}(\varepsilon) \;\approx\; \tfrac{1}{2}\mathrm{Const}_p(\varepsilon) - \tfrac{1}{2}\big(\mathrm{Main}_p(\varepsilon) + \Gamma_p(\varepsilon) + \mathrm{Err}(\varepsilon)\big),$$

so $\mathrm{Const}^+_{p,N}$ is not a half-normalisation of Const_p; the two differ by the half-sum of the other terms. The proxy ratio $r^+_{p,N}(\varepsilon) = |\mathrm{Const}^+_{p,N}|/|\mathrm{Main}_p|$ approaches a value $\le \frac{1}{2}$ as $\varepsilon \to 0$ (Remark 4.2) because $|\mathrm{Main}_p| \sim \varepsilon^{-1}$ dominates the $O(1)$ difference term in the limit. No exact identity with Const_p of Proposition 2.2 is claimed without the bridge. Representative values are listed below.

p	$\mathrm{Main}_p(0.05)$	$\Gamma_p(0.05)$	$\mathrm{Const}^+_{p,N}(0.05)$
2	-2.77	-0.475	2.44
7	-4.15	-0.193	3.47
23	-3.69	-0.105	3.79
37	-3.35	-0.082	3.93
53	-3.08	-0.069	3.74

At $\varepsilon = 0.05$, the pointwise inequality $Z^+_{p,N}(0.05) < 0$ holds for 14 of the 16 primes $p \le 53$. The exceptions are $p = 37$ and $p = 53$, for which $Z^+_{p,N}(0.05) \approx +0.50$ and $+0.59$ respectively. This is *not* a failure of the underlying decomposition: it reflects the finite-ε maximum of the ratio $r(\varepsilon)$ near $\varepsilon \approx 0.05$, which narrowly exceeds unity for exactly these two primes. At the tested value $\varepsilon = 0.020$ the pointwise bias holds for all 16 primes without exception. At $\varepsilon = 0.025$, only $p = 53$ fails the pointwise negativity condition ($Z^+_{53,N}(0.025) = +0.389 > 0$), while $p = 37$ remains negative ($Z^+_{37,N}(0.025) = -0.356 < 0$). At $\varepsilon = 0.030$, both exceptions $p = 37, 53$ are present. The sign-crossover for $p = 53$ lies in (0.020, 0.025), and that for $p = 37$ lies in (0.025, 0.030). Whether the bias persists continuously throughout $\varepsilon \in (0, 0.020]$ is not established here.

No structural explanation for the exceptions $p = 37, 53$ at $\varepsilon = 0.05$ is currently available. The phenomenon appears to be a finite-ε, finite-κ manifestation of the observed unimodal peak of $r(\varepsilon)$ near $\varepsilon = 0.050$, which for exactly these two primes narrowly exceeds unity. Whether this reflects an underlying arithmetic structure or is a numerical coincidence at $\kappa = 53$ is not addressed here.

The observed negativity of $Z^{+}_{p,N}(\varepsilon)$ should be read neither as a prime race in the sense of Rubinstein–Sarnak [19] nor as a Landau–Gonek prime-power asymptotic; it is a different, prime-indexed sign statistic for Gaussian-smoothed explicit-formula evaluations.

Status. The tested values of $r^{+}_{p,N}(\varepsilon)$ are numerically consistent with decrease toward a value satisfying $r^{\infty}_{p} \le \frac{1}{2}$ (Remark 4.2). The sign of the ordinate proxy $Z^{+}_{p,N}$ remains the subject of Problem 8.2. Establishing a uniform bound on the finite-ε correction, which would close the asymptotic bias conjecture in its sharpest form, is part of Problem 8.2 of § 8. The empirical sign-crossover localisation above, at $\kappa = 53$, is the sharpest statement that the present methods allow.

Remark (Relation to Balanzario–Cárdenas Romero)**.** *Balanzario and Cárdenas Romero [24] study a Gaussian-Hermite-weighted explicit formula* $S(\xi, k) = (1/\xi)\sqrt{k/2\pi}\,\sum_n \exp\{-2(k/\xi^2)(\gamma_n - \xi)^2\}$ *with a negative prime-side leading term of the form* $-c_2\,\Lambda(n)/\sqrt{n}\cdot\exp\{-(\xi^2/4k)\log^2 n\}$*, which is structurally analogous to our* $\mathrm{Main}_p(\varepsilon) = -\log p/(2\sqrt{\pi}\,\varepsilon\sqrt{p})$*. Their parametrisation uses Hermite polynomials and prime-power sums* $\Lambda(n)$*; our single-prime, leading-*ε *form with explicit constant* $1/(2\sqrt{\pi})$ *appears to be a new observation.*

6 The Integrated Bias

The two expressions for B used below are the same quantity in two representations: the primal sum

$$B \;=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p)$$

and the dual sum $B = \sum_p (\log p)^2\,\mathrm{Re}\,Z_p^{(2)}$. The equivalence between the two is the Curvature-Bias Decomposition (cf. [5], §5) and is treated here as imported. All numerical values in this section refer to the same object.

We use the dual representation as a starting point. The curvature bias at $\sigma = \frac{1}{2}$ is encoded in the weighted sum

$$B \;:=\; \sum_{p \le \kappa} (\log p)^2\,\mathrm{Re}\,Z_p^{(2)},$$

which was introduced in [5] as $-\frac{1}{2}$ times the $\sigma = \frac{1}{2}$ evaluation of the second σ-derivative of the trace identity, via the identity $O''(\frac{1}{2}) = -2B$. Its structural decomposition is algebraic and does not depend on any unproved hypothesis.

Proposition 6.1 (Curvature–bias identity; proved)**.** *Let* $O(\sigma) = \frac{1}{2}\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\cos(2\sigma\gamma_k \log p)$

and define

$$B \;:=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p).$$

Then

$$O''(\tfrac{1}{2}) \;=\; -2\,B.$$

Proof. Direct differentiation gives

$$O'(\sigma) \;=\; -\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,\gamma_k \log p\,\sin(2\sigma\gamma_k \log p),$$

$$O''(\sigma) \;=\; -2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(2\sigma\gamma_k \log p).$$

Evaluation at $\sigma = \frac{1}{2}$ yields

$$O''(\tfrac{1}{2}) \;=\; -2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p) \;=\; -2\,B. \qquad \square$$

Remark. *This identity shows that all curvature information at $\sigma = \frac{1}{2}$ is encoded in a single weighted cosine interaction between primes and zeros.*

Numerically, at reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, the direct curvature sum equals $B = -19\,342.5$, hence $O''(\frac{1}{2}) = -2B = +38\,685$. This value coincides with $\mathrm{Tr}(L)$, where $L = -d^2\widetilde{T}(\sigma)/d\sigma^2\big|_{\sigma=1/2}$ is the curvature operator of [5], and provides an independent closed-form cross-check for the identity above.

Remark. *The quantity B is defined as in [5, §5], and the identity $O''(\frac{1}{2}) = -2B$ follows from the explicit form of $O(\sigma)$ given there. The numerical value $B = -19\,342.5$ at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ is the one quoted throughout the series.*

Theorem 6.2 (Integrated proxy bias is negative; numerical)**.** *At $(\kappa, \varepsilon, N) = (53, 0.05, 100)$,*

$$B^+_{\mathrm{int}}(0.05, 100) \;:=\; \sum_{p \le 53} (\log p)^2\, Z^+_{p,N}(0.05) \;=\; -42.21 \;<\; 0.$$

Observation 6.3 (Direct curvature sum; numerical)**.** *At $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $B = -19\,342.5 < 0$, equivalently $O''(\frac{1}{2}) = -2B = +38\,685 > 0$.*

When the parameters $(\varepsilon, N) = (0.05, 100)$ are fixed throughout, we write B_{int} as shorthand for $B^+_{\mathrm{int}}(0.05, 100)$.

Remark. *The values B and B_{int} refer to two* different *quantities, not to a single quantity in two forms. B is the direct curvature sum weighted by $(\gamma_k \log p)^2$, obtained by differentiating the trace identity and evaluating at $\sigma = \frac{1}{2}$; $B^+_{\mathrm{int}}(\varepsilon, N)$ is the integrated proxy bias, weighted only by $(\log p)^2$ and summed over $Z^+_{p,N}(\varepsilon)$. The two differ by a reweighting through the γ_k^2-factors absent in the second. Both are numerically negative at the reference parameters, and the shared negative sign is a numerical finding rather than the consequence of a proved equivalence.*

Proposition 6.1 connects B — not B_{int} — to the curvature $O''(\frac{1}{2})$. Whether pointwise negativity of $Z^{+}_{p,N}(\varepsilon)$, or the integrated proxy bias $B^{+}_{\mathrm{int}}(\varepsilon, N) < 0$, implies the corresponding negativity of B at the same parameters is an open question; it is formulated as Problem 8.4 of § 8.

Remark. *Although $Z^{+}_{p,N}(0.05) > 0$ at $p = 37$ and $p = 53$, the $(\log p)^2$-weighted sum B_{int} remains decisively negative: the negative contributions from the remaining 14 primes dominate the positive contributions from these two. Concretely, $(\log 37)^2 \cdot 0.50 + (\log 53)^2 \cdot 0.59 \approx 6.52 + 9.30 = 15.82$, against a cumulative negative contribution of -58.03 from the other primes, yielding the net value -42.21. The integrated bias is robust against individual-prime deviations as long as these remain isolated.*

7 Positive Curvature at the Critical Line

We now record the curvature statement that follows from Observation 6.3 and Proposition 6.1, and, separately, the conditional strict-local-extremum statement that follows once stationarity is added as a hypothesis.

Corollary 7.1 (Positive curvature at $\sigma = \frac{1}{2}$; numerical)**.** *At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, the direct curvature sum B is numerically negative (Observation 6.3), $B = -19\,342.5 < 0$. By Proposition 6.1, this is equivalent to*

$$O''(\tfrac{1}{2}) \;=\; -2B \;=\; +38\,685 \;>\; 0.$$

Proof. By Observation 6.3, $B = -19\,342.5$. By Proposition 6.1, $O''(\frac{1}{2}) = -2B$. □

Remark. *If, in addition to $B < 0$ at the reference parameters (here numerically established), the stationarity condition $O'(\frac{1}{2}) = 0$ holds at the same parameters, then $\sigma = \frac{1}{2}$ is a strict local minimum of O, and the trace*

$$\operatorname{Tr}\widetilde{T}(\sigma) \;=\; D_{\mathrm{SEL}} - O(\sigma)$$

attains a strict local maximum at $\sigma = \frac{1}{2}$.

The exact vanishing condition $O'(\frac{1}{2}) = 0$ is stronger than the first-derivative cancellation bound formulated in Problem 8.1 and remains open. Numerically, $O'(\frac{1}{2}) = +2.475$ is approximately 4% of the natural scale $W_{\varepsilon,N} \cdot P(\kappa)$, consistent with the cancellation expected from the odd kernel $\sin(\gamma_k \log p)$, but no analytical bound on $|O'(\frac{1}{2})|$ is established in the present paper.

Remark. *The paper [5] recorded four further numerical signatures concentrating at $\sigma = \frac{1}{2}$: a $+47\%$ trace spike relative to the σ-neighbourhood average, the dominance of the leading eigenvalue μ_1, the minimum of the spectral gap, and the joint maximum of $\mathrm{Tr}(L)$ and $\lambda_{\max}(L)$. Each of these is numerically robust and independent of the bias conditions established here. They constitute additional numerical signatures from [5] at the same reference parameters, but do not enter the proof of Corollary 7.1 or of the conditional remark above.*

Remark. *The curvature statement identifies $\sigma = \frac{1}{2}$ as a geometrically distinguished point of the spectral trace at the reference parameters. It is a structural ingredient in a possible route to Weil* *positivity in the Lagarias formulation [7] (Problem 8.5 of § 8), but does not by itself establish it. No implication concerning the Riemann Hypothesis is asserted or implied by Corollary 7.1, nor by the conditional remark above when the stationarity hypothesis is imposed. The curvature identity therefore provides a canonical second-order signature of the critical line, independent of global hypotheses.*

8 Open Problems

The present work reduces the question of curvature at the critical line to five well-posed subproblems. We name each and describe the tools we expect to be relevant. The problems are listed in order of what we consider tractability.

Open Problem 8.1 (First-derivative cancellation and localisation)**.** *The trivial triangle inequality gives*

$$|O'_{(\kappa,\varepsilon,N)}(\tfrac{1}{2})| \;\le\; P(\kappa)\sum_{k=1}^{N}\gamma_k e^{-\varepsilon^2\gamma_k^2} \;=\; \widetilde{C}_{\varepsilon,N}\,W_{\varepsilon,N}P(\kappa), \qquad \widetilde{C}_{\varepsilon,N} := \frac{\sum_{k=1}^{N}\gamma_k e^{-\varepsilon^2\gamma_k^2}}{\sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}},$$

which is finite for fixed (ε, N) but does not represent cancellation. The nontrivial problem is to prove a first-derivative cancellation estimate *of the form: for some $C_0 > 0$ and $A \ge 1$,*

$$\big|S_\kappa(\gamma)\big| \;\le\; C_0\,\frac{P(\kappa)}{\gamma(\log\gamma)^A}, \qquad S_\kappa(\gamma) := \sum_{p\le\kappa}(\log p)\sin(\gamma\log p),$$

uniformly in $\kappa \ge 2$ and $\gamma \ge \gamma_1$. This would imply

$$|O'_{(\kappa,\varepsilon,N)}(\tfrac{1}{2})| \;\le\; \frac{C_0\,W_{\varepsilon,N}\,P(\kappa)}{(\log\gamma_1)^A},$$

which is a genuine logarithmic saving over the trivial bound. The numerical value at reference parameters is $O'(\frac{1}{2}) = +2.475$, approximately 4% of $W_{\varepsilon,N}\cdot P(\kappa)$, consistent with the cancellation expected from the odd kernel $\sin(\gamma_k \log p)$. The expected tools are: equidistribution of the sequence $\{\gamma_k \log p \bmod 2\pi\}$ under appropriate averaging; a discrete Weyl-type bound exploiting the independent behaviour of zero heights and prime logarithms. At the reference parameters, $B < 0$ is numerically established (Corollary 7.1); the remaining open hypothesis is the prime-sum cancellation estimate $|S_\kappa(\gamma)| \le C_0P(\kappa)/(\gamma(\log\gamma)^A)$ stated above. Such a bound would become a localisation statement only when combined with a local convexity lower bound and a smallness condition relative to that lower bound, as detailed below. Specifically, if there exist $\delta > 0$ and $m > 0$ such that $O''(\sigma) \ge m$ on $[\frac{1}{2}-\delta, \frac{1}{2}+\delta]$ and $|O'(\frac{1}{2})| < m\delta$, then there exists $\sigma_* \in [\frac{1}{2}-\delta, \frac{1}{2}+\delta]$ with $O'(\sigma_*) = 0$ and

$$|\sigma_* - \tfrac{1}{2}| \;\le\; \frac{|O'(\frac{1}{2})|}{m}.$$

The reference-parameter estimate gives $\sigma_* - \frac{1}{2} \approx -O'(\frac{1}{2})/O''(\frac{1}{2}) \approx -6.4\times10^{-5}$. An exact local minimum at $\sigma = \frac{1}{2}$ requires additionally $O'(\frac{1}{2}) = 0$; this stronger condition remains open.

Remark (Possible conditional route to stationarity)**.** *A possible partial route toward Problem 8.1 would be a cancellation estimate for the prime exponential sum*

$$S_\kappa(\gamma) := \sum_{p\le\kappa} (\log p)\sin(\gamma\log p).$$

Since

$$O'(\tfrac{1}{2}) = -\sum_k e^{-\varepsilon^2\gamma_k^2}\gamma_k\, S_\kappa(\gamma_k),$$

an estimate of the form

$$|S_\kappa(\gamma)| \le C_0\,\frac{P(\kappa)}{\gamma(\log\gamma)^A}, \qquad P(\kappa) := \sum_{p\le\kappa}\log p,$$

for constants $C_0 > 0$ and $A \ge 1$ (uniform in $\kappa \ge 2$ and $\gamma \ge \gamma_1$) would imply

$$|O'(\tfrac{1}{2})| \;\le\; C_0 P(\kappa)\sum_k \frac{e^{-\varepsilon^2\gamma_k^2}}{(\log\gamma_k)^A} \;\le\; \frac{C_0\,W_{\varepsilon,N}\,P(\kappa)}{(\log\gamma_1)^A}.$$

No such estimate is proved here. The hypothesis is analogous to Vinogradov–Korobov-type cancellation ([26, 27], [28], [25]) and is strictly weaker than RH. The structural obstruction to an unconditional proof is described in Remark 8. Numerically, $O'(\frac{1}{2}) = +2.475$ and $W_{\varepsilon,N}\cdot P(\kappa)/\log\gamma_1 \approx 23.3$ at the reference parameters $(\varepsilon, N) = (0.05, 100)$, consistent with the route above. The route falls short of an unconditional proof: the logarithmic saving required to establish $|S_\kappa(\gamma)| \le C_0P(\kappa)/(\gamma(\log\gamma)^A)$ uniformly in κ is not available; see Remark 8.

Remark (Why non-vanishing on $\Re s = 1$ is insufficient)**.** *Direct unconditional proofs of the stationarity bound face a structural obstruction: the function $-\zeta'/\zeta(s - i\gamma)$ has a pole at $s = 1 + i\gamma$, so the non-vanishing $\zeta(1 + it) \ne 0$ (Hadamard [10], de la Vallée Poussin) alone does not prevent cancellation failure in the sum $\sum_{p\le\kappa}(\log p)\sin(\gamma\log p)$. Thus non-vanishing on $\Re s = 1$ alone is insufficient for this route. A cancellation estimate of the strength displayed in Remark 8, or an alternative estimate of comparable strength, is still required for this route.*

Open Problem 8.2 (Asymptotic ordinate bias and finite-N transfer)**.** (a) Asymptotic target. *Let $Z_p^\infty(\varepsilon) := 2\sum_{k=1}^\infty e^{-\varepsilon^2\gamma_k^2}\cos(\gamma_k\log p)$ be the infinite ordinate proxy. Prove analytically that $Z_p^\infty(\varepsilon) < 0$ for every prime $p \le \kappa$ and all sufficiently small $\varepsilon > 0$.*

(b) Finite-N transfer. *For specified (N, ε), establish $|R_{p,N}^\infty(\varepsilon)| < |Z_p^\infty(\varepsilon)|$ uniformly for $p \le \kappa$, where $R_{p,N}^\infty(\varepsilon) := 2Z_{p,N}^+(\varepsilon) - Z_p^\infty(\varepsilon)$ is the tail beyond the N-th zero. Then $Z_{p,N}^+(\varepsilon) < 0$ follows.*

The numerical behaviour of $r_{p,N}^+(\varepsilon)$ (Remark 4.2) suggests the expected sign for the leading asymptotic of the full GW object, but the limit $r_p^\infty \le \frac{1}{2}$ is not proved analytically; the corresponding sign of Z_p^∞ is not directly implied (the GW bridge remains open; see Paper 5, Open Problem 6.5 (GW bridge)). The open question is an analytic proof that $Z_p^\infty(\varepsilon) < 0$ for all primes and small but positive ε. Numerically, pointwise negativity of $Z_{p,N}^+$ is established on the tested grid at $\varepsilon = 0.020$ for all primes $p \le 53$. Closing this gap would prove the asymptotic pointwise-bias target in part (a) and, together with the finite-N transfer in part (b), would give a theorem-level sign statement for $Z_{p,N}^+$ in the corresponding small-ε regime. It would not *by itself* *prove the finite-grid unimodality pattern or the observed grid transition of Theorem 5.1; those remain numerical unless separately analysed. The expected tools are: explicit expansion of the $s = 0, 1$ endpoint terms in the completed Guinand–Weil formula with the correct* cosh*-evaluation of the test function; a Riemann–Lebesgue-type bound on the digamma contribution with explicit constant; a cancellation argument between the ε^{-1}-coefficients of* Main_p *and* Const_p*. The problem is formulated entirely within classical analytic number theory.*

Open Problem 8.3 (Uniformity of positive curvature)**.** *Determine whether the inequality $B(\varepsilon, \kappa) < 0$, and hence $O''(\frac{1}{2}) > 0$, persists at parameter points (κ, ε, N) other than the reference values $(53, 0.05, 100)$ tested here. Corollary 7.1 establishes the inequality at a single point of the parameter space. Whether it extends to a neighbourhood, to a larger range of cutoffs κ, or to a different choice of smoothing ε is a distinct question: the direct curvature sum depends non-trivially on the γ_k^2-weights and on the oscillatory phases $\gamma_k \log p$, and finite parameter changes could in principle reverse the sign. The expected tools are: a systematic numerical survey across a parameter grid; an asymptotic expansion of $B(\varepsilon, \kappa)$ for large κ; and, if possible, an analytic reduction via the primal-to-dual identity $B = \sum_p (\log p)^2 \operatorname{Re} Z_p^{(2)}$.*

Open Problem 8.4 (Integrated-to-direct transfer)**.** *Establish or disprove the implication*

$$\big(Z^+_{p,N}(\varepsilon) < 0 \textit{ for all primes } p \le \kappa \big) \implies B(\varepsilon, \kappa) < 0$$

at fixed (ε, κ), where

$$B(\varepsilon, \kappa) \;:=\; \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \, (\gamma_k \log p)^2 \, \cos(\gamma_k \log p).$$

At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, B^+_{int} and B are numerically negative (Theorem 6.2, Corollary 7.1), while the pointwise inequality $Z^+_{p,N}(0.05) < 0$ holds for 14 of the 16 primes $p \le 53$, with exceptions $p = 37, 53$ (Theorem 5.1). On the tested grid at $\varepsilon = 0.020$, the pointwise inequality holds for all primes $p \le 53$. No analytical implication among these sign statements is established. A proof of the displayed implication would extend Corollary 7.1 from a reference-parameter statement to a parametrised statement valid wherever the pointwise hypothesis can be verified. The expected tools are: the explicit prime-side representation of B, the uniform positivity of the Fourier transform $\widehat{h}_{p,\varepsilon}$, and a rearrangement argument distributing the γ_k^2-weights across the pointwise contributions. The analogous implication with B^+_{int} in place of B is immediate from the positivity of the weights $(\log p)^2$: if $Z^+_{p,N}(\varepsilon) < 0$ for all $p \le \kappa$, then $B^+_{\mathrm{int}} = \sum_p (\log p)^2 Z^+_{p,N} < 0$. The nontrivial question is whether any sign information survives the additional γ_k^2-reweighting in B. The problem is formulated entirely within classical analytic number theory.

Open Problem 8.5 (Lagarias–Weil bridge)**.** *The present result introduces a finite-cutoff curvature observable for Gaussian-smoothed explicit-formula data; it should be compared with Bombieri's finite truncations of Weil's quadratic functional [16] and with the Li/Bombieri–Lagarias positivity framework [17, 18], but it is not yet a positivity theorem for the full Weil distribution.*

Problem 8.5 asks whether there exists a transfer construction under which the finite-cutoff curvature condition $O''(\frac{1}{2}) = -2B > 0$, together with stationarity $O'(\frac{1}{2}) = 0$ and an appropriate limiting procedure, yields Weil positivity for a specified Gaussian-generated subclass of Lagarias-admissible test functions (entire of order 2, explicit-formula admissibility conditions). This problem is stated here as a named research target, not as a proved implication. Establishing this route precisely would require specifying the admissible test-function class, constructing an explicit transfer from the finite-parameter curvature identity $O''(\frac{1}{2}) = -2B$ to the limiting Weil functional, and proving positivity on that limiting class; none of these steps is supplied in the present paper. Its resolution is beyond the methods of the present paper and requires either a direct positivity argument on the Weil distribution or an external route via Connes-type operator positivity. We are not aware of a direct transfer theorem connecting the curvature statement $O''(\frac{1}{2}) > 0$ or the Lagarias formulation of Weil positivity [7] to Bombieri's quadratic functional [16] or the Connes–Consani Weil-positivity framework [21]. The two bodies of work remain disjoint as of the date of this paper; Problem 8.5 is therefore genuinely open. No implication toward the Riemann Hypothesis is asserted by the present paper; the problem above is a named research target, not an established claim.

Problems 8.1–8.4 are formulated entirely within classical analytic number theory; the present problem is structurally the principal outstanding question of the programme.

Remark (Outlook: the Weil transfer operator)**.** *The curvature identity $O''(\frac{1}{2}) = -2B$ established in this paper admits a natural continuation: a possible construction of a finite-dimensional Weil transfer matrix $W_{\kappa,\varepsilon}$ whose entries would be evaluations of the Weil quadratic form on the admissible Gaussian test functions $h^{\text{even}}_{p,\varepsilon}$. The diagonal of such a matrix would be connected to the curvature sum B via the identity $O''(\frac{1}{2}) = -2B$, while its off-diagonal structure encodes the prime-zero coupling studied in Papers 1–5 of this series.*

The Hilbert space H_W derived from the Weil hermitian form (see Suzuki [29]) is the natural infinite-dimensional ambient space in which the finite-dimensional matrix $W_{\kappa,\varepsilon}$ lives as a finite-rank object. Connes and Consani [21] construct the matrix $\sigma(n,m) = Q_W(\eta_n, \eta_m)$ of the Weil quadratic form on a basis $\{\eta_n\}$ indexed by integers, using prolate-spheroidal cutoffs on a compact interval $[1/\lambda, \lambda]$. A possible future construction of the Weil transfer operator $W_{\kappa,\varepsilon}$ is indexed by primes $p \leq \kappa$ rather than integers n, uses Gaussian smoothing of width ε on the spectral side rather than a prolate-spheroidal cutoff, and is connected to the curvature identity $O''(\frac{1}{2}) = -2B$ established here via its diagonal. The two constructions are methodologically related but technically distinct.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The ordinates γ_k are used as numerically specified inputs, and no property of their distribution requiring the Riemann

Hypothesis is assumed or used.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The operator $\widetilde{T}$ and its σ-parametrised extension $\widetilde{T}(\sigma)$ are constructed explicitly from Gaussian-smoothed prime phasors in the previous papers of this series [3, 4, 5]. No random-matrix hypothesis, no pair-correlation conjecture, and no Hilbert–Pólya postulate enters the construction or the proofs.

$\widetilde{T}$ is not a Hilbert–Pólya operator. As observed numerically in [5] on the tested finite grids, the eigenvalues follow an inverse-ordinate pattern $\mu_j \approx C_T/\gamma_{k(j)}$, which is incompatible with a Hilbert–Pólya interpretation on those numerical diagnostics. The paper does not attempt a Hilbert–Pólya realisation.

The Guinand–Weil formula is classical. Proposition 2.2 is a classical identity valid for every even, sufficiently rapidly decaying test function; its derivation does not require the truth of the Riemann Hypothesis. Theorems 3.1 and 4.2, together with the algebraic identity of Proposition 6.1, are proved unconditionally by direct computation. Theorem 4.3 verifies numerically that the $N = 100$ truncation tail is negligible at the stated reference parameters.

Numerical results are clearly marked. Theorems 4.1, 5.1, and 6.2, and Corollary 7.1, rely on numerical evaluation of finite sums or one-dimensional numerical quadrature; where zero-ordinate truncation enters, the tail is controlled by Theorem 4.3. Each such statement is labelled [numerical] in its heading.

Conditional statements are explicitly flagged. The conditional remark following Corollary 7.1 is stated conditionally on the stronger exact stationarity condition $O'(\frac{1}{2}) = 0$, which is distinguished from the weaker first-derivative cancellation target in Problem 8.1, and adds no further unproved input beyond this explicit assumption. Problem 8.4 is named openly and is not used as a hypothesis anywhere in §§ 3–7.

γ_k as inputs, not conclusions. The ordinates appear as numerically specified inputs to the model. The claim $\mathrm{Re}(\rho_k) = \frac{1}{2}$ is not used anywhere; the reference point $\sigma_0 = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

The author welcomes critical feedback, corrections, and collaboration: Zenodo (`https://zenodo.org/search?q=Ulrich+Tehrani`) and GitHub (`https://github.com/utehrani/`

`analysislab-nt`).

Topics of particular interest include: analytical approaches to the asymptotic pointwise bias of Problem 8.2; the analytic stationarity bound of Problem 8.1; uniformity of the curvature inequality across a larger parameter range (Problem 8.3); the integrated-to-direct transfer of Problem 8.4; and, ultimately, the Lagarias–Weil bridge of Problem 8.5.

We particularly welcome expert comment on the following question: for a Gaussian-smoothed family g_ε with $g_\varepsilon \to \delta$ as $\varepsilon \to 0$, is it known whether the sign of $W(g_\varepsilon * g_\varepsilon^\vee)$ at finite ε carries arithmetic information, or is positivity only meaningful in the distributional limit?

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